

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA5115

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

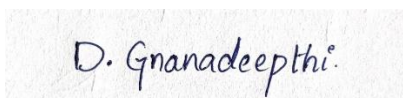
Comparative Analysis

Code of Conduct:

I D. Gnana Deepthi (192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



D. Gnanadeepthi.

Annexure A

Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.
2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.
3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.
4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.
5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

ANSWER

3. Comparative Analysis of PGP and S/MIME

Pretty Good Privacy (PGP) and Secure/Multipurpose Internet Mail Extensions (S/MIME) are widely used standards for securing email communication. Both provide confidentiality, integrity, authentication, and non-repudiation, but they differ in key management, trust models, scalability, and enterprise usability.

Aspect	PGP	S/MIME
Key Management	Users generate and manage their own public/private key pairs. Public keys are commonly shared through key servers.	Uses X.509 digital certificates issued and managed by Certificate Authorities (CAs).
Trust Model	Uses a decentralized Web of Trust, where users verify and sign each other's public keys.	Uses a centralized hierarchical trust model based on trusted CAs.
Scalability	Can become difficult to manage in large organizations because users are responsible for key verification and trust relationships.	Highly scalable because certificates can be centrally issued, renewed, revoked, and managed.
Enterprise Usability	More suitable for individual users, privacy-focused communities, and decentralized environments.	Better suited for organizations because it integrates easily with corporate email systems and centralized security policies.

Aspect	PGP	S/MIME
Administration	Requires more user involvement in managing keys and verifying identities.	Can be centrally administered by an IT security team.
Email Integration	Supported by various email clients, often requiring additional configuration.	Widely supported by enterprise email clients such as Microsoft Outlook.

Recommendation for a Corporate Email System

For a corporate email system, S/MIME is the most suitable approach. Its centralized certificate management, hierarchical trust model, scalability, and easier integration with enterprise email infrastructure make it more practical for large organizations. IT administrators can control certificate issuance, renewal, revocation, and security policies efficiently.

Conclusion:

While PGP provides strong security and flexibility through its decentralized Web of Trust, S/MIME is recommended for corporate environments due to its better scalability, centralized management, and enterprise usability.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand